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E 231 CABLE TRAY SYSTEM

1. SCOPE OF WORK

The Works shall include the design, supply, manufacturing, factory testing, delivery to Site, installation, commissioning and testing of the cable tray system in accordance with Part 4, Subsection 1.2 Summary of Work.

The cable trays, covers and structures also included for 150 kV Power Cable connection to the 150 kV Substation from generator transformer.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following specific standards:

2.1 National

- PUIL : Peraturan Umum Instalasi Listrik - Indonesia
(General Regulation of Electrical Installation Indonesia).
- SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 American Society for Testing and Materials

- ASTM A36 : Standard Specification for Carbon Structural Steel
- ASTM B633 : Specification for Electrodeposited Coatings of Zinc on Steel
- ASTM A123 : Specifications for Zinc Coating (Hot-Dip) on Assembled Steel Products
- ASTM A653 : Specifications for General Requirements for Steel Sheet, Zinc Coated (Galvanized) by the Hot-Dip Process.

2.3 American Society of Mechanical Engineers

- ASME : Boiler and Pressure Vessel Code, Section IX Welding and Brazing Qualifications

2.4 American Welding Society

- ANSI/AWS D1.1 : Structural Welding Code-Steel

2.5 National Electrical Manufacturers Association

NEMA VE1 : Metal Cable Tray Systems.

3. DESIGN REQUIREMENTS

Separate cable trays shall be used for High Voltage, Medium Voltage and Low Voltage power, control and instrumentation cables.

Low voltage power and control cables may be run in the same cable tray with the Owner's written approval in special cases.

Segregated cable tray runs shall be provided for each of the redundant power distribution systems and associated control and instrumentation.

Tray fill shall have allowance for future installation of 10% more cables. Cable tray supports shall be designed with a safety factor of not less than two (2).

Two-hour rated fire stops shall be installed in vertical and horizontal cable trays where they penetrate floors or walls.

Cable tray and accessories shall be based on types that are manufactured in Indonesia or equivalent.

4. DETAILED REQUIREMENTS

4.1 General

All cable trays for power and control cables shall be ladder bottom type with standardized width of 300 or 450 or 600 mm. Perforated cable trays 150 mm wide may be used for a maximum of three (3) low voltage power cables or control cables. The width of the cable trays shall not exceed 600 mm. The tray depth shall not exceed 100 mm. All cable trays for instrumentation cables shall be ladder bottom tray or solid-bottom or perforated type, 100 or 150 mm wide, with tight, fitting covers in accordance with the instrument systems requirement.

Vertical clearances between the cable trays shall not be less than 350 mm, lower edge to lower edge. The access way between the cable trays or between a wall and the cable trays shall not be less than 600 mm.

The top of the highest cable tray shall have at least 350 mm clearance from the ceiling, beams, or overhead structures.

Covers shall be provided on ladder-type cable trays at places where falling objects may damage the cables, and at places advised by the Owner.

Cable trays shall preferably be located above process piping and facilities.

All hardware shall be made from suitable corrosion-proof materials.

4.2 Loading Requirements

Cable trays and supports shall be designed for the following minimum loading requirements.

- (1) Design load capability, with supports on 3.5 meter centers shall be not less than 280 kg/m² with a safety factor of 2 when loaded as a simple beam in accordance with NEMA VE1 or equivalent.
- (2) Maximum deflection at mid-span for loading stated above with simple beam support shall be 1/300 of the span of cable trays.
- (3) Trays and fittings shall be capable of supporting a concentrated static load with simple beam supports on 3.5 meters on centers without permanent deformation, as follows:
 - a. 90 kg applied vertically over a 100 mm longitudinal section at mid-span of one side member over a connector, with tray loaded to design load, or
 - b. 90 kg applied vertically over a 100 mm section at mid-point of any rung of an empty ladder type section or a mid-point of empty solid bottom section. Mid-point is defined as midway between side members.
- (4) The wind load for cable trays mounted outdoors shall be based on a wind velocity of 120 km/h.

4.3 Construction

4.3.1 Cable Trays

Cable trays shall be made up in sections, fittings, and accessories of the ladder and solid-bottom type as defined in NEMA VE1.

All cable tray sections, verticals, covers and fittings shall be fabricated from a minimum of 2 mm thick sheet steel. Tray components, except as otherwise noted herein, shall be hot-dipped galvanized after fabrication in accordance with ASTM A123. Covers shall be hot-dipped galvanized.

The average thickness of the coating shall be 64 micrometers and the minimum thickness not less than 55 micrometers.

Cable tray covers shall be of the raised, flanged type, except for instrument trays which shall be tight fitting to provide a continuous steel envelope around the cables. Covers for cable trays 600 mm and wider shall be provided with angle or channel stiffeners to resist deflection when loaded.

The solid bottom or perforated cable trays shall be provided with stiffeners at the bottom, if necessary, to prevent deflection relative to the side rails. The maximum deflection with design uniform loading shall not exceed 5 mm.

The cable tray sides shall be flanged outward with minimum width of 25 mm. Ends of cable tray sections and fittings shall be fabricated for bolted connections with splice plates.

Splice plates, channels, stiffener bars, etc. shall be steel in accordance with ASTM A 36 Specifications.

Curved plates for cable protection shall be provided at angles, corners, etc. Curved plates shall be welded at the top and bottom, on both sides.

After manufacturing of cable trays and fittings is completed, the pieces shall be inspected for damage and all welds, cuts, marred surfaces, etc. shall be repaired and galvanized with a suitable galvanizing touch up material.

4.3.2 Cable Tray Supports

Cable tray supports for suspended cable trays shall consist of continuously threaded (rolled thread) rod, hexagon nuts, hex stock coupling with running thread and steel structural members (angle and/or channel).

The rods, nuts and couplings shall be hot-dipped galvanized in accordance with ASTM A 123. The steel structural members shall be hot-dipped galvanized after fabrication in accordance with ASTM A123

Rods shall be minimum 1.27 mm diameter. All steel structural members shall comply with ASTM A36.

4.4 Installation

The Bidder shall assemble and install all the cable tray sections, fittings, support hangers, brackets, miscellaneous hardware and the required grounding to the structural steel to form a complete and electrically continuous cable tray system. Rods shall be attached to anchors with support rod couplings.

It shall be the Bidder's responsibility to ensure that all rods are threaded into couplings and inserts a minimum of 13 mm.

Cable tray covers shall be installed on all instrument trays, all cable trays installed outdoors and on all trays under or within two (2) meters of a gallery or tray crossing. Covers shall be fastened with corrosion-proof hexhead self-tapping screws. Cable tray risers shall have covers up to 200 cm above the floor level.

Where cable trays are installed directly over equipment, such as MCCs, switchgear, etc., the vertical clearance between the top of equipment and bottom of the cable trays shall be not less than 600mm, where possible.

5. PERFORMANCE TESTING AND GUARANTEES

Type test reports shall be submitted for the following items:

- (1) Cable tray loading tests as described in Clause 4.2 above.
- (2) Fire rating and hose stream tests for fire stops.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for cable tray system shall conform to the requirements included in Part 4 section 2, Subsection 2.7. The documentation as specified shall include, but not be limited to, the followings:

- (1) Test reports for tray loading and deflection for each type.
- (2) Electrical continuity test reports.
- (3) All documentation required by applicable codes and standards.

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